



Apr 12, 2025

Rabat, Morocco

Modern Applications of Cryptography

Youssef El Housni | yelhousni.eth

Consensys
New York City | United States

Linea[•]

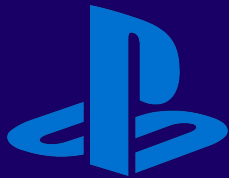


Youssef El Housni



- PhD | Ecole Polytechnique Paris
- Cryptographer | Consensys NYC
- Co-founder of gnark
- Co-founder of Linea

Linea

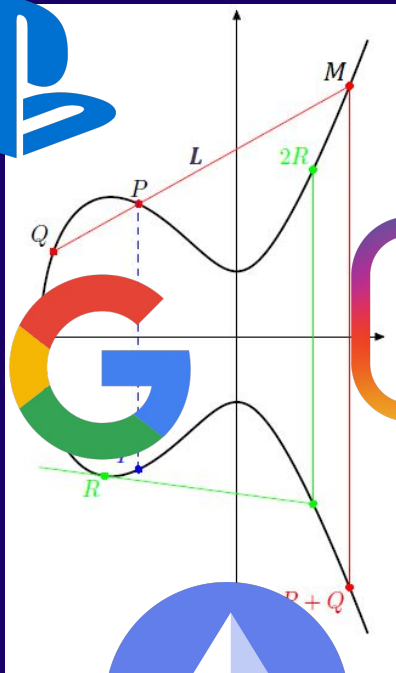
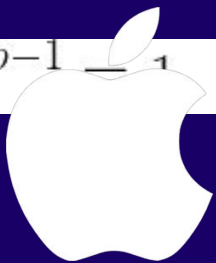


Linea

$$f(x) = a_0 + a_1x + \cdots + x_n x^n, a_i \in \mathbb{Z}/p\mathbb{Z}$$



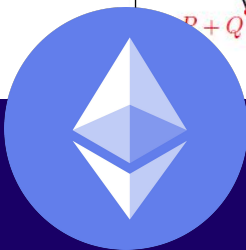
$$x^{p-1} \equiv 1 \pmod{p}$$



$$A_{m \times n} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$



$$\mathbb{P}(X) \quad \mathbb{P}(X) \quad \mathbb{P}\left(\frac{X}{2}\right)$$
$$\mathbb{P}(X|Y) \quad \mathbb{P}(X|Y)$$



$$n \quad p_2^{n_2} \cdots p_k^{n_k} = \prod_{i=1}^k p_i^{n_i}$$

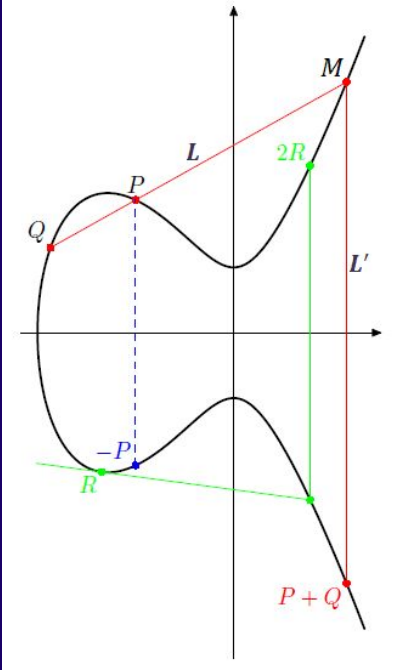


Linea

$$f(x) = a_0 + a_1x + \cdots + x_nx^n, a_i \in \mathbb{Z}/p\mathbb{Z}$$

$$x^{p-1} \equiv 1 \pmod{p}$$

$$\begin{aligned} &\mathbb{P}(X) \quad \mathbb{P}(X) \quad \mathbb{P}\left(\frac{X}{2}\right) \\ &\mathbb{P}(X \mid Y) \quad \mathbb{P}(X \mid Y) \quad \mathbb{P}\left(\frac{X}{2} \mid Y\right) \end{aligned}$$



$$A_{m \times n} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

$$n = p_1^{n_1} \cdot p_2^{n_2} \cdots p_k^{n_k} = \prod_{i=1}^k p_i^{n_i}$$

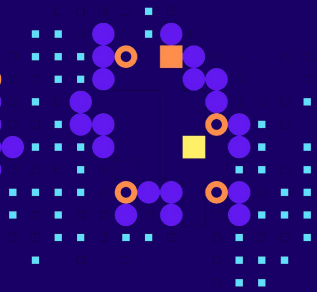
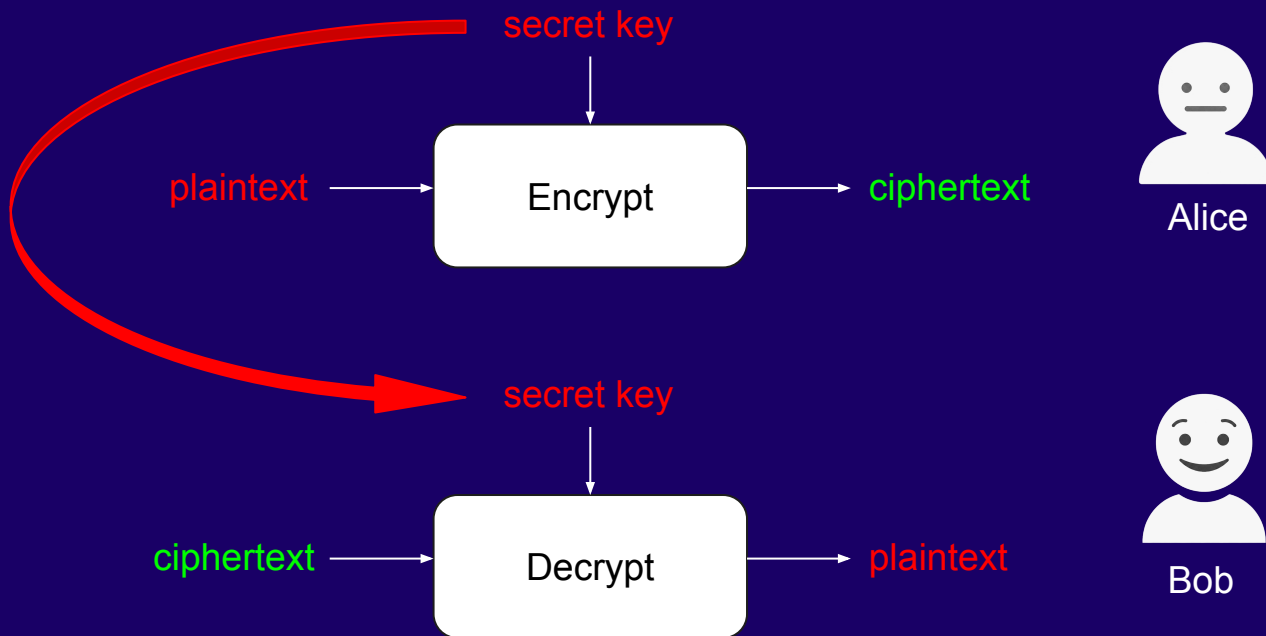
Crypt-o-graphy

- The prefix “crypt-” means hidden
- The suffix “-graphy” means writing

So, all together it says “hidden writing”



Encryption



Encryption



Hieroglyphs



[Caesar Cipher](#)

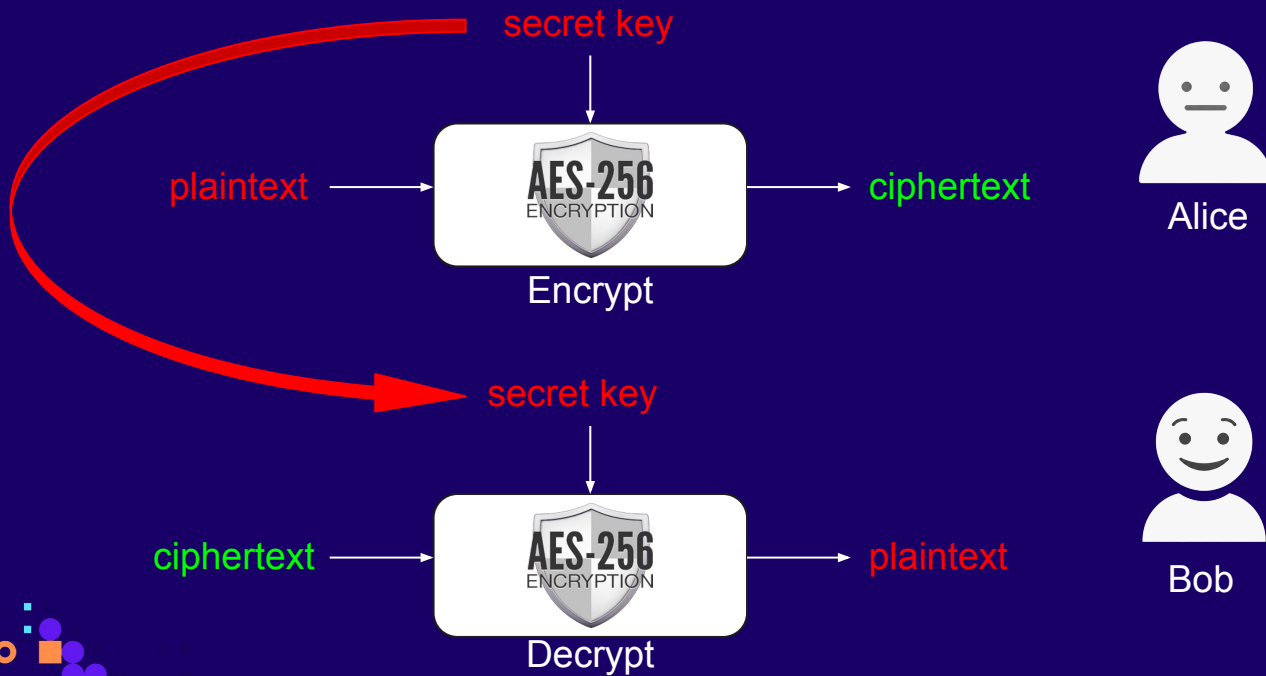


[Current standard](#)

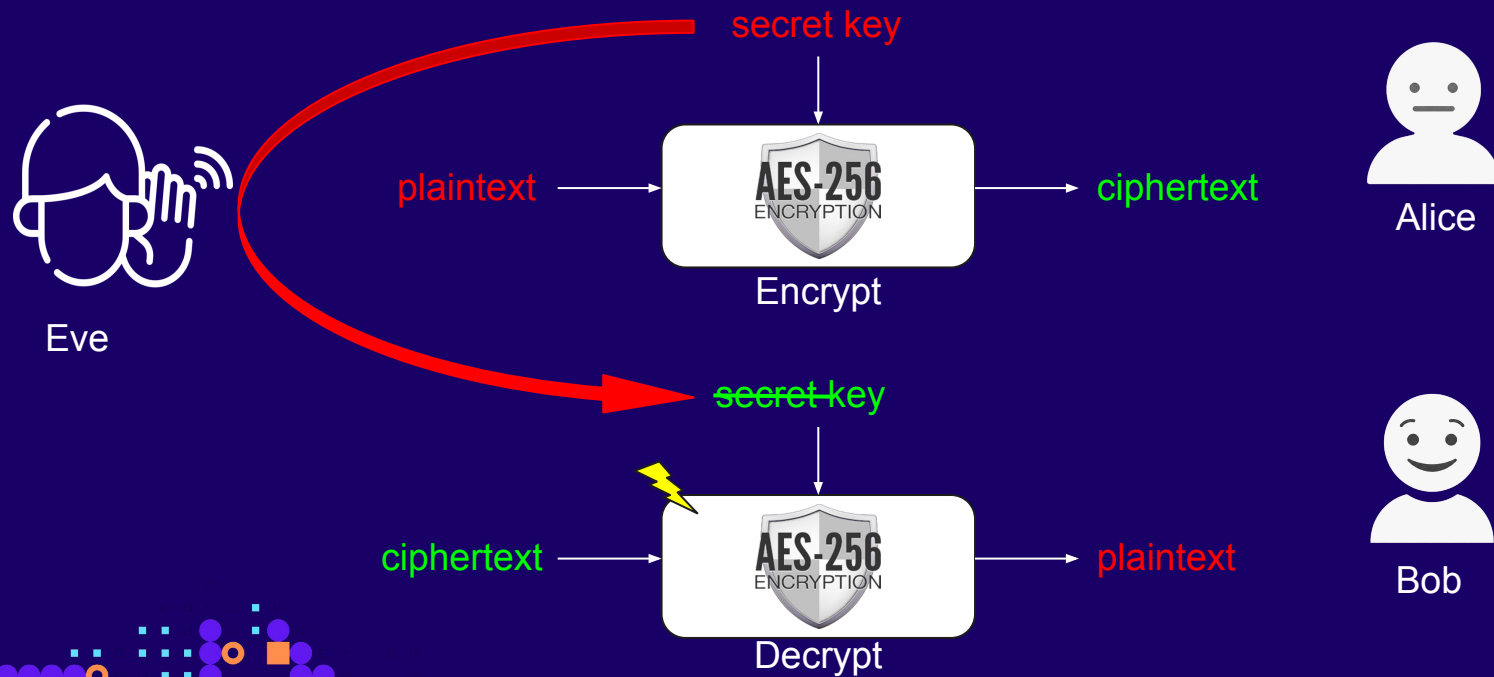


[Enigma WWII](#)

Encryption



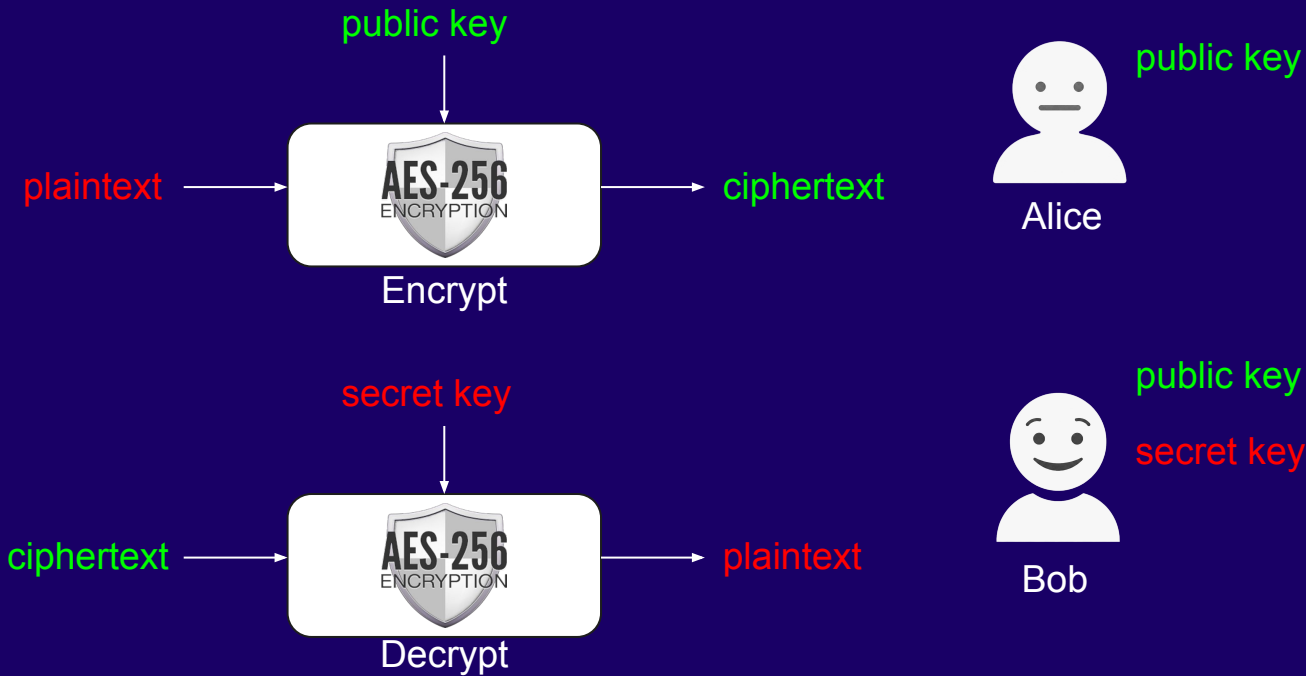
Encryption



Public key encryption



(secret) key $\longrightarrow$ (public) key
 $\longleftarrow$



(secret) key $\longrightarrow$ (public) key
 $\longleftarrow \text{no}$

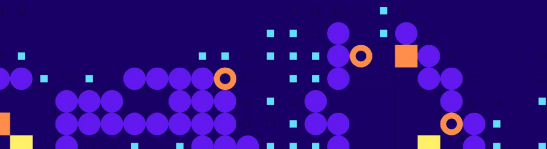
Trapdoor function

$$n = p_1^{n_1} \cdot p_2^{n_2} \cdots p_k^{n_k} = \prod_{i=1}^k p_i^{n_i}$$

$$p_1, p_2 \longrightarrow n = p_1 \cdot p_2$$

$$n = p_1 \cdot p_2 \xrightarrow{\text{no}} p_1, p_2$$

$$6895601 = 1931 \times 3571$$



Computational complexity

n	2^n	Examples
32	$2^{32} = 10^{9.6}$	number of humans on Earth
47	$2^{47} = 10^{14.2}$	distance Earth - Sun in millimeters ($149.6 \cdot 10^{12}$) number of operations in one day on a processor at 2 GHz
56	$2^{55.8} = 10^{16.8}$	number of operations in one year on a processor at 2 GHz
79	$2^{79} = 10^{23.8}$	Avogadro number: atoms of Carbon 12 in 1 mol
82	$2^{82.3} = 10^{24.8}$	mass of Earth in kilogrammes
100	$2^{100} = 10^{30}$	number of operations in $13.77 \cdot 10^9$ years (age of the universe) on a processor at 2 GHz
155	$2^{155} = 10^{46.7}$	number of molecules of water on Earth
256	$2^{256} = 10^{77.1}$	number of electrons in universe

RSA Encryption

1977, Rivest, Shamir, Adleman

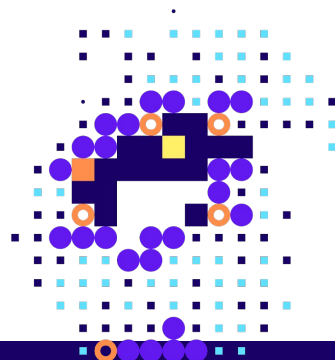
- modulus $N = p \times q$, p, q two distinct large primes
- arithmetic modulo N , in $\mathbb{Z}/N\mathbb{Z} = \{0, 1, \dots, N - 1\}$

The **multiplicative group** is the set of **invertible** integers in $\{1, 2, \dots, N - 1\}$.
invertible x means $\gcd(x, N) = 1$, x coprime to N .

There are $\varphi(N) = (p - 1)(q - 1)$ invertible integers in $\{1, \dots, N - 1\}$

Hard tasks without knowing p, q if N is large enough:

- computing $(p - 1)(q - 1)$,
- computing a square root $\sqrt{x} = x^{1/2} \bmod N$,
- computing an e -th root $x^{1/e} \bmod N$.



End-to-end encryption



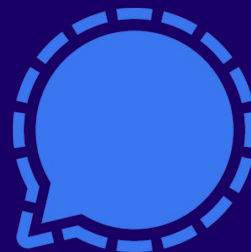
- By default
- Closed-source



- By default
- Closed-source



- By default
- Open-source client
- closed-source server



- By default
- Open-source client
- Open-source server

(secret) key $\longrightarrow$ (public) key
 $\longleftarrow$ 

Signature



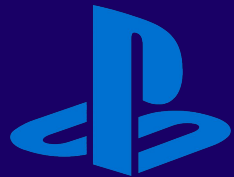
$$s \cdot G \rightarrow_{\text{easy}} P$$

$$P(= s \cdot G) \rightarrow_{\text{hard}} s$$



public key P

Alice



- Only Bob can sign with s
- Alice (or anyone) can verify with P

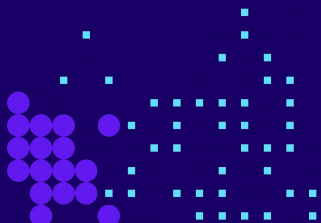


public key P

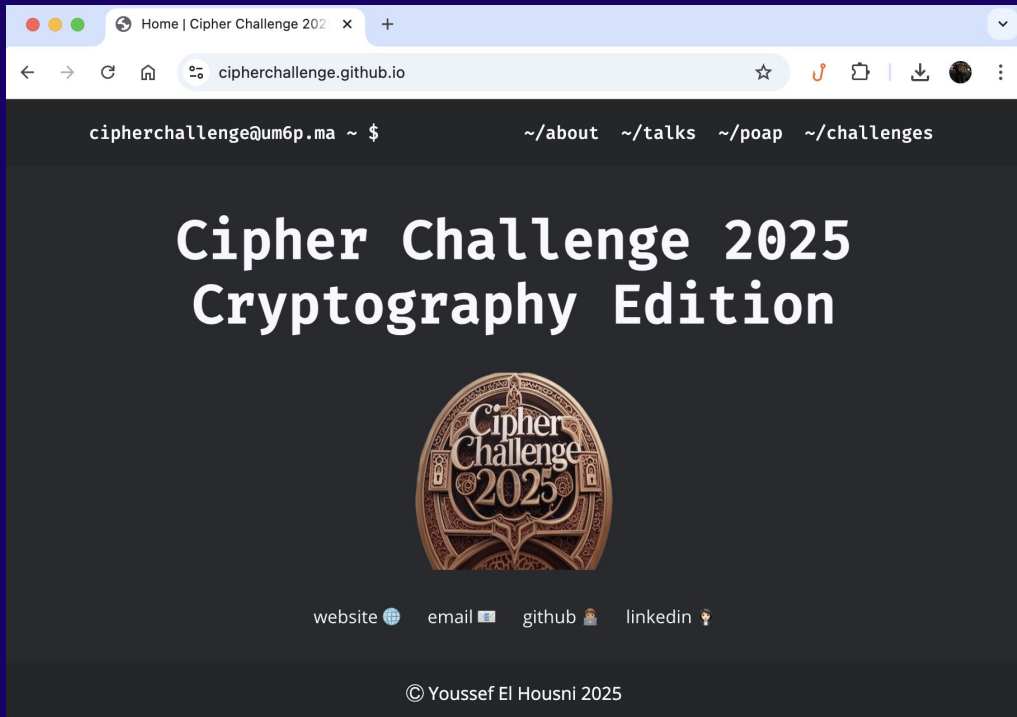
secret key s

Bob

$$x^{p-1} \equiv 1 \pmod{p}$$

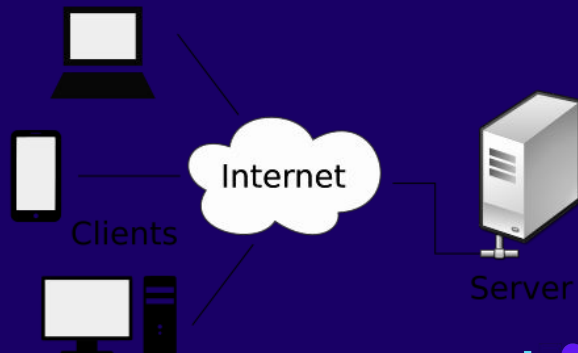


Traditional cryptography



<https://cipherchallenge.github.io>

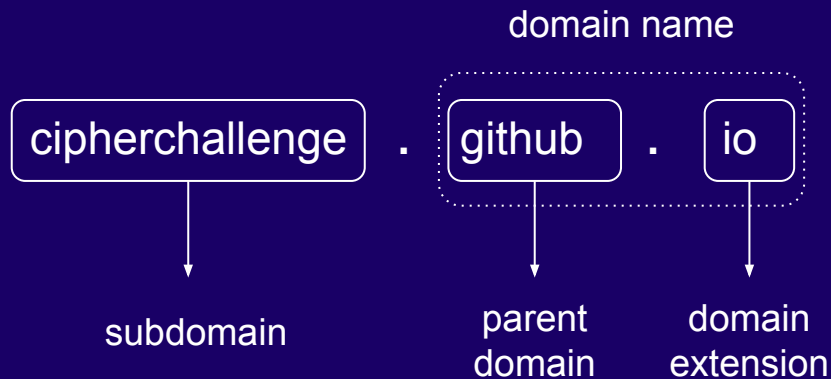
TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256, 128 bits, TLS 1.2



Websites



- Hosting (server)
- Domain name resolution (DNS)



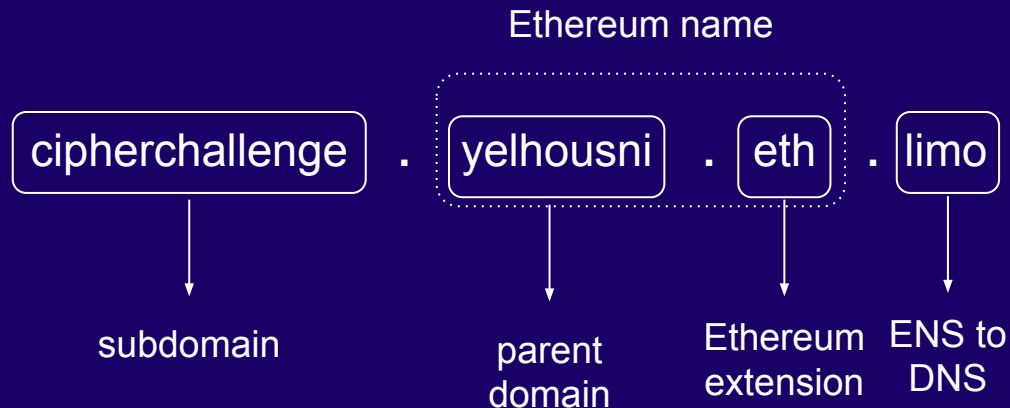
IP addresses for **cipherchallenge.github.io**

Our DNS servers responded with these IP addresses when we queried it for the domain cipherchallenge.github.io. Some DNS servers may return different IP addresses based on your location.

IP address	Type	Hosted by	Location
> 185.199.108.153	IPv4	Fastly, Inc.	United States of America

Decentralized Websites

- Decentralized Hosting (p2p IPFS)
- Ethereum name resolution (ENS)



 `bafybeiclwibybzcxnmbudh5xekmvwhgkhfiod7q4arfimhszru6vhcpjhi.ipfs.dweb.link`

Modern cryptography

$$f(x) = a_0 + a_1x + \dots + x_nx^n, a_i \in \mathbb{Z}/p\mathbb{Z}$$

Secret sharing:

Secret key s :

$s = s_1, s_2, s_3, \dots, s_n$

$$s = a_0$$

$$s_i = f(i)$$

E.g. DNSSEC

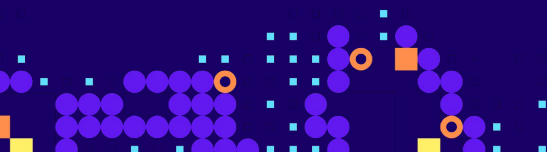
root key s

$s = s_1, \dots, s_7$

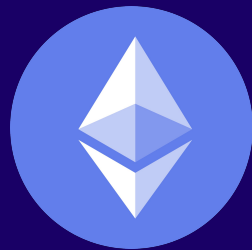
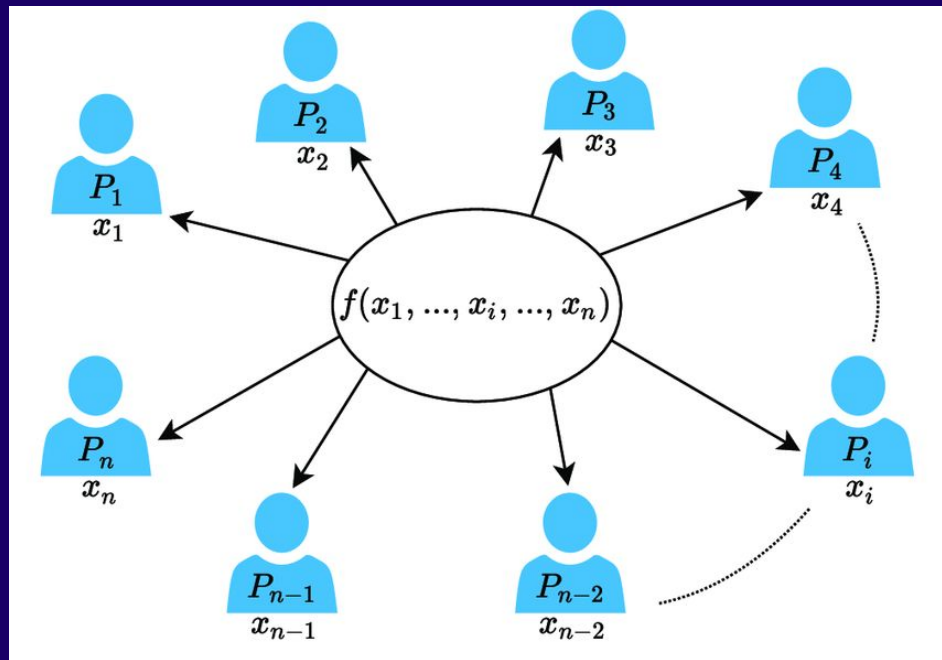
5/7 keys to restart the internet

$$f(1), \dots, f(n) \rightarrow_{\text{find}} f(x)$$

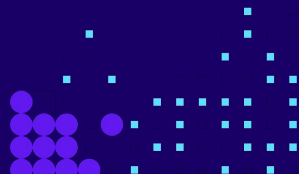
$$s = f(0)$$



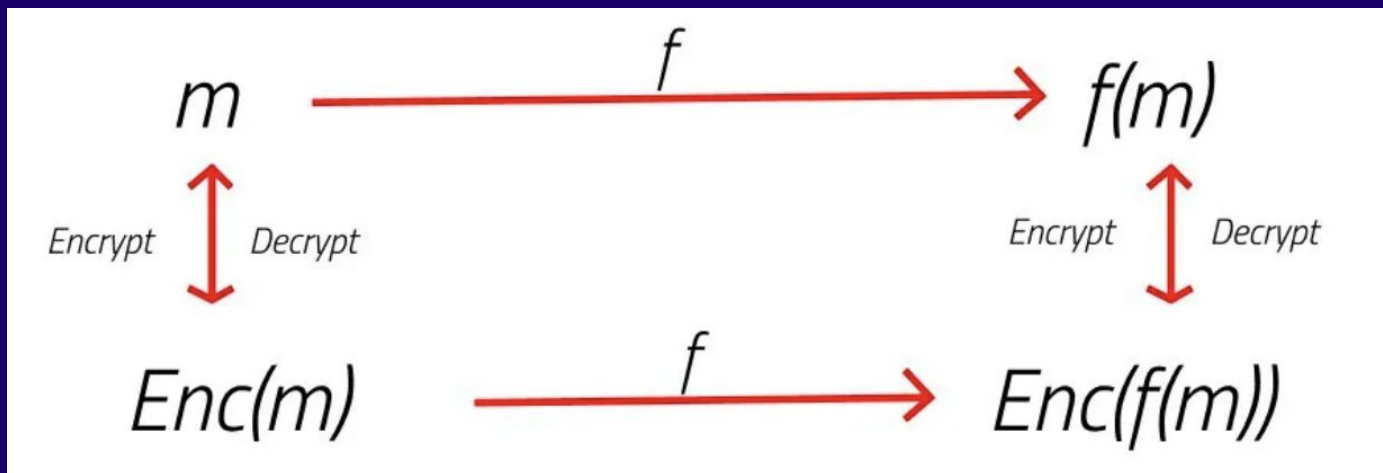
Multi-party computation



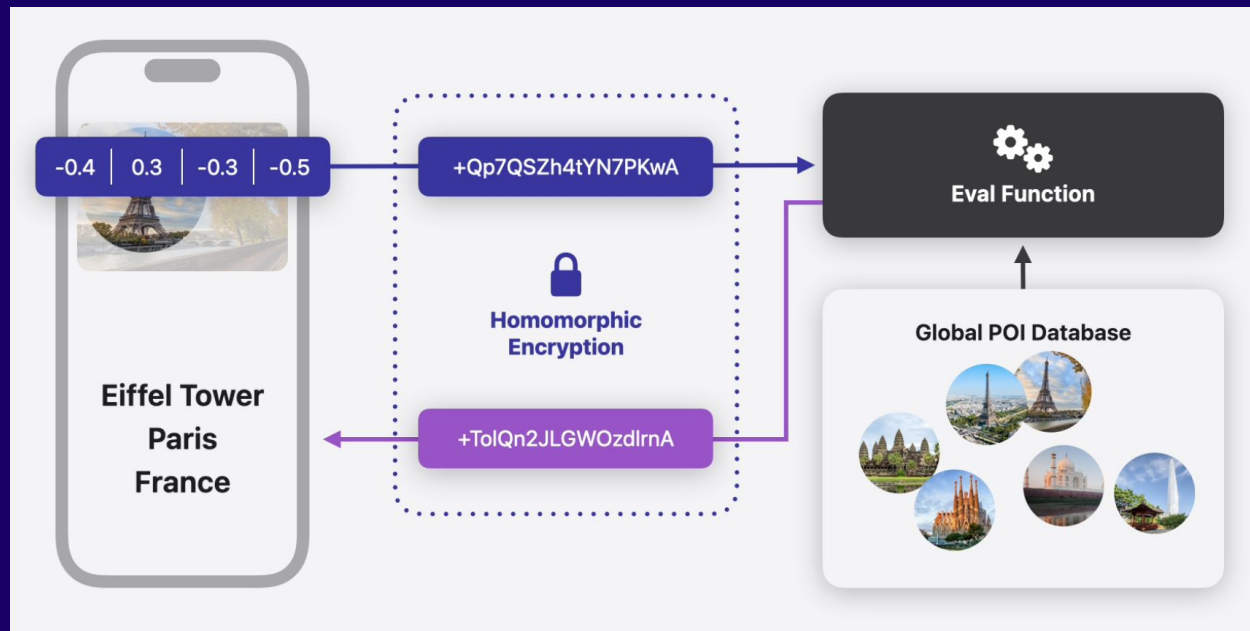
N people sign (collaboratively) a transaction



Fully Homomorphic Encryption



Apple's Private Nearest Neighbor Search

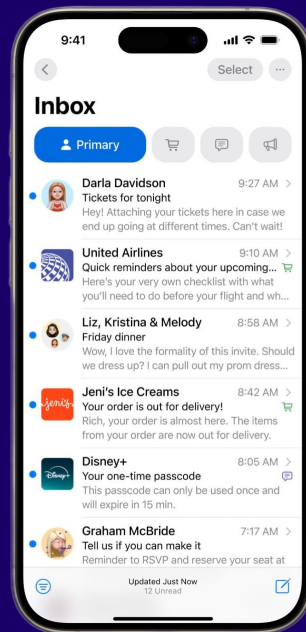


<https://machinelearning.apple.com/research/homomorphic-encryption>

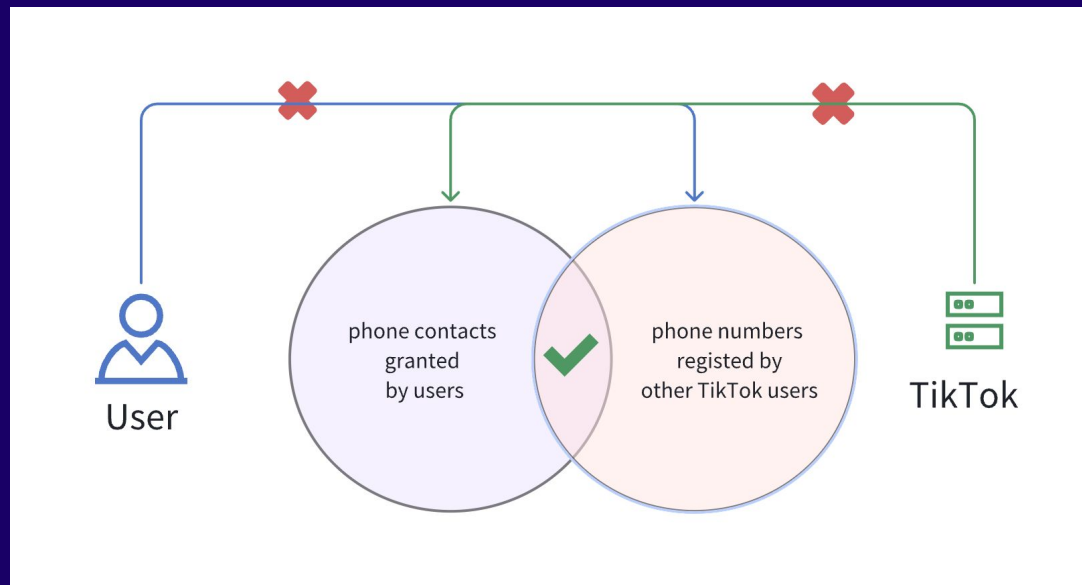
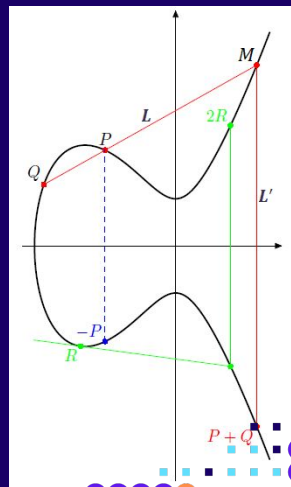
Apple's Private Information Retrieval



$$A_{m \times n} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$



<https://machinelearning.apple.com/research/homomorphic-encryption>



<https://developers.tiktok.com/blog/tiktok-practices-in-privacy-enhancing-technologies>

Zero-knowledge proofs

Alice

I know the solution to
this complex equation

Bob

No idea what the solution is
but Alice claims to know it

Challenge



Response



- **Sound:** **Alice** has a **wrong solution** $\implies$ **Bob** is **not convinced**.
- **Complete:** **Alice** has the **solution** $\implies$ **Bob** is **convinced**.
- **Zero-knowledge:** **Bob** does NOT learn the solution.

Linea

$$f(x) = a_0 + a_1x + \cdots + x_nx^n, a_i \in \mathbb{Z}/p\mathbb{Z}$$

Zero-knowledge proofs

$$A_{m \times n} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$



I have a **zk proof** that
X sent **\$Z** to **Y**



Linea

I have a **short proof** that
the transaction X is correct

|transaction| >> |proof|

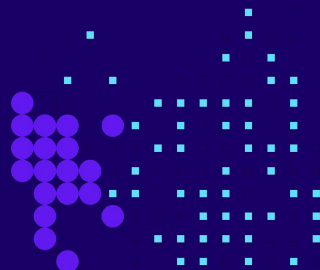
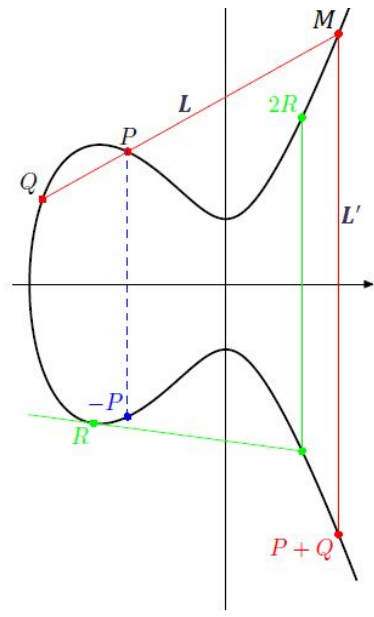
<https://z.cash>

<https://www.getmonero.org>

<https://ethereum.org>

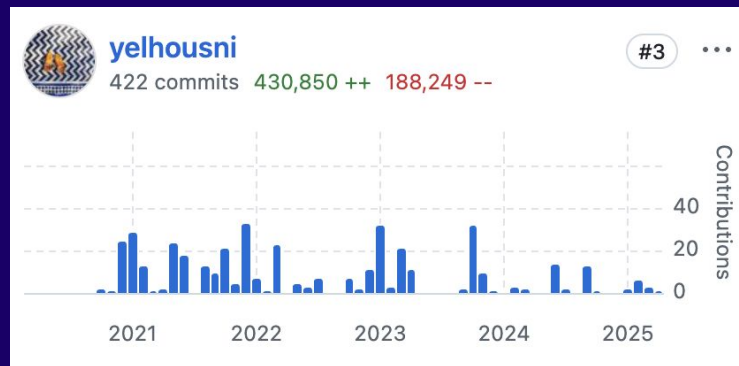
<https://linea.build>

$$\begin{array}{ccc} \mathbb{P}(X) & \mathbb{P}(X) & \mathbb{P}\left(\frac{X}{2}\right) \\ \mathbb{P}(X | Y) & \mathbb{P}(X | Y) & \mathbb{P}\left(\frac{X}{2} | Y\right) \end{array}$$



Zero-knowledge proofs

- Free Cryptography library: <https://github.com/Consensys/gnark-crypto>
- Free ZKP library: <https://github.com/Consensys/gnark>
- Playground: <https://play.gnark.io/>
 - Factorisation example: <http://play.gnark.io/?id=petqlbhyng>



EdMSM: Multi-Scalar-Multiplication for SNARKs and Faster Montgomery multiplication

Gautam Botrel and Youssef El Housni

Linea, ConsenSys
gautam.botrel@consensys.net

Families of prime-order endomorphism-equipped embedded curves on pairing-friendly curves

Antonio Sanso¹ and Youssef El Housni²

¹ Ethereum Foundation
² Linea

Pairings in Rank-1 Constraint Systems

Youssef El Housni^{1,2,3[0000-0003-2873-3479]}

¹ ConsenSys R&D, gnark team, Paris, France
² LIX, CNRS, École Polytechnique, Institut Polytechnique de Paris
³ Inria

Optimized and secure pairing-friendly elliptic curves suitable for one layer proof composition

Youssef El Housni^{1,2,3[0000-0003-2873-3479]} and Aurore Guillevic^{4[0000-0002-0824-7273]}

¹ EY Blockchain, Paris, France
² LIX, CNRS, École Polytechnique, Institut Polytechnique de Paris
³ INRIA
youssef.el.housni@fr.ey.com
⁴ Université de Lorraine, CNRS, Inria, LORIA, Nancy, France
aurore.guillevic@inria.fr

A survey of elliptic curves for proof systems*

Diego F. Aranha^{1[0000-0002-2457-0783]},
Youssef El Housni^{2,3,4[0000-0003-2873-3479]}, and
Aurore Guillevic^{1,5[0000-0002-0824-7273]}

¹ Aarhus University, Aarhus, Denmark
dfaranha@cs.au.dk
² ConsenSys, gnark, Paris, France
³ LIX, CNRS, École Polytechnique, Institut Polytechnique de Paris
⁴ Inria
youssef.elhousni@consensys.net
⁵ Université de Lorraine, CNRS, Inria, LORIA, Nancy, France
aurore.guillevic@inria.fr

Co-factor clearing and subgroup membership testing on pairing-friendly curves*

Youssef El Housni^{1,2,3[0000-0003-2873-3479]},
Aurore Guillevic^{4,5[0000-0002-0824-7273]}, and
Thomas Piellard¹

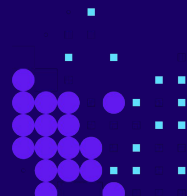
¹ ConsenSys, gnark
youssef.elhousni@consensys.net

Families of SNARK-friendly 2-chains of elliptic curves*

Youssef El Housni^{1,2,3[0000-0003-2873-3479]}
and Aurore Guillevic^{4,5[0000-0002-0824-7273]}

¹ ConsenSys, gnark, Paris, France
² LIX, CNRS, École Polytechnique, Institut Polytechnique de Paris
³ Inria
youssef.elhousni@consensys.net
⁴ Université de Lorraine, CNRS, Inria, LORIA, Nancy, France
⁵ Aarhus University, Aarhus, Denmark
aurore.guillevic@inria.fr

École Polytechnique,
Paris, France



Cryptanalysis

a.k.a. Attacks

- [Underlying math](#)
- [Implementation](#)
- [Side-channel](#)
- [Social/frontend](#)
- [Political](#)
- [Future](#)

An efficient key recovery attack on SIDH

Wouter Castryck^{1,2} and Thomas Decru¹

¹ imec-COSIC, KU Leuven, Belgium

² Vakgroep Wiskunde: Algebra en Meetkunde, Universiteit Gent, Belgium

27th Chaos Communication Congress

Console Hacking 2010

PS3 Epic Fail

fail0verflow

bushing, marcan, segher, sven

North Korean hackers cash out hundreds of millions from \$1.5bn ByBit hack

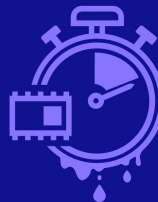
10 March 2025

Share Save

Apple pulls data protection tool after UK government security row

22 February 2025

Share Save

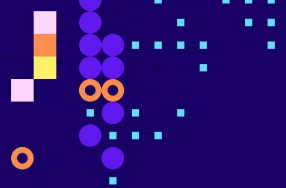


TPM-Fail

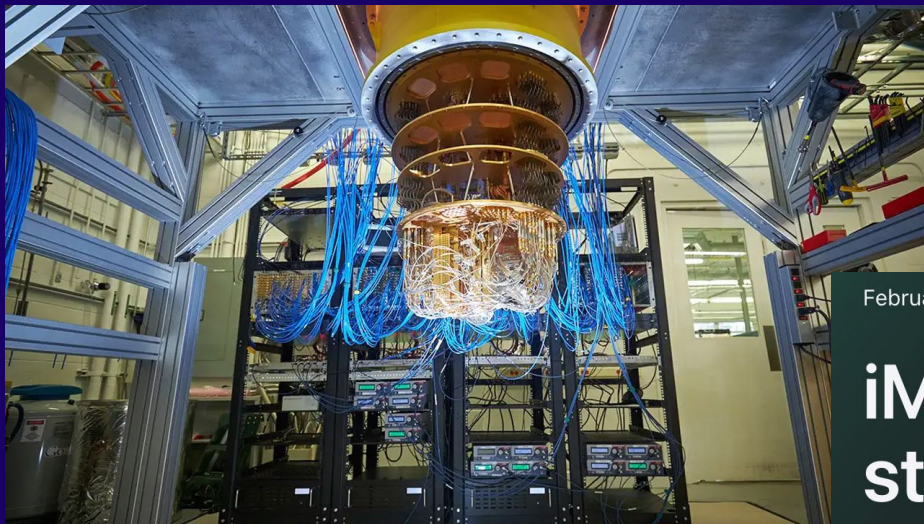
TPM MEETS TIMING AND LATTICE ATTACKS

DOWNLOAD PAPER

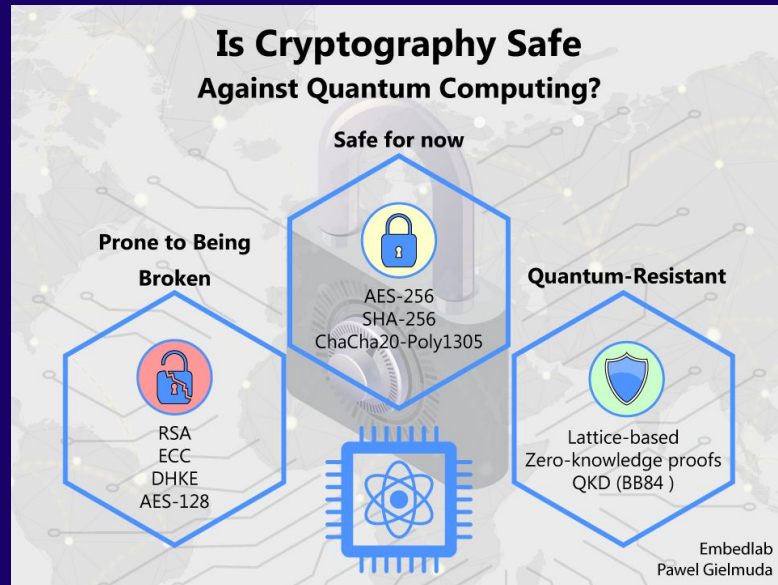
CITE PAPER



Future attacks



Quantum computer



February 21, 2024

iMessage with PQ3: The new state of the art in quantum-secure messaging at scale

Posted by Apple Security Engineering and Architecture (SEAR)

<https://security.apple.com/blog/imessage-pq3/>

Proof of Attendance Protocol (POAP)



YOU GOT A POAP!



Cipher Challenge 2025 - UMP6 CC

📅 Apr 12, 2025 - Apr 13, 2025
📍 Rabat, Morocco

This POAP proves you have attended the UMP6P CC Cipher Challenge in Rabat, 12-13 April, 2025.

cipherchallenge.github.io

Collect this POAP

Mint now

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Thank you

yelhousni.eth.limo
linea.build
gnark.io

youssef.elhousni@consensys.net

Telegram: @ElMarroqui

GitHub: @yelhousni

X: @YoussefElHousn3

